

Assignment 2 — Covariance, VaR, and Copulas

FinTech 545 — Quantitative Risk Management

Puhan Yang

Table of contents

1	Conventions	1
2	Correlations from Mismatched Histories	3
2.1	Predict	3
2.2	Fit	4
2.3	Reconcile	6
3	A Volatility Estimate After the Regime Changed	8
3.1	Predict	8
3.2	Fit	9
3.3	Reconcile	10
4	When Diversification Raises VaR	12
4.1	Predict	12
4.2	Fit	13
4.3	Reconcile	13
5	Gaussian or t Copula	16
5.1	Predict	16
5.2	Fit	17
5.3	Reconcile	19
6	Model Based Simulation and Residual Correlation	21
6.1	Predict	21
6.2	Fit	21
6.3	Reconcile	22

1 Conventions

Every number quoted here comes from the output of the script named at the head of its section, and every figure is written by the same script. `README.md` gives the commands, and `output/` holds the saved console output.

- **VaR and ES** are reported as positive numbers for a loss, at $\alpha = 5\%$ unless stated. A negative VaR means the α quantile of the P&L is still a profit. The historical quantile follows the class implementation (`library/RiskStats.jl`): sort the P&L and average the order statistics at $\lfloor n\alpha \rfloor$ and $\lceil n\alpha \rceil$. ES is minus the mean of every P&L at or below that point.
- **Moments**. Variance with the $n - 1$ divisor; skewness and kurtosis bias corrected; kurtosis in *excess* form, so a normal gives 0.
- **Returns** are arithmetic, from `riskmgmt.return_calculate`.
- **Exponentially weighted variance** is `riskmgmt.ew_covar`, the class estimator: weights $(1 - \lambda)\lambda^{i-1}$ normalized to sum to 1, and the *weighted* mean removed before squaring.
- **AICc** counts every fitted parameter: $k = 2$ for a normal margin, $k = 3$ for a t margin. For the copulas the margins are frozen first and only the copula's own parameters count: $k = 0$ for the Gaussian, $k = 1$ (ν) for the t , with R held at the same Kendall estimate for both (Week 05). $\text{BIC} = k \ln m - 2\ell$.
- **Simulations** use fixed seeds and 100,000 draws. Where two models are compared, they share their random numbers, so the difference between them is the model and not the draw. Where a comparison is close, its simulation standard error is estimated from ten batches of 10,000.

The Week 03 machinery comes from the repository's `riskmgmt` library, which passes functional tests 1.1–7.6. Problem 2 turned up a bug in its `fit_general_t`: on de-meaned data the location started at about 10^{-19} , scipy's Nelder–Mead stepped it by 5% of that, and the location never moved. It returned $\nu = 8.36$ where the true MLE is 8.27. The fix is a separate commit and all 25 tests still pass.

2 Correlations from Mismatched Histories

Code: *problem1.py*.

2.1 Predict

Table 1: Days each pair of series is jointly observed

	A	B	C	D	IDX
A	242	242	221	93	242
B		242	221	93	242
C			228	83	228
D				96	96
IDX					250

All five trade on 81 of the 250 days, and every one of those days is from day 151 on, because D lists on day 151 (Figure 1). C misses 22 days on its foreign calendar, A and B miss 8.

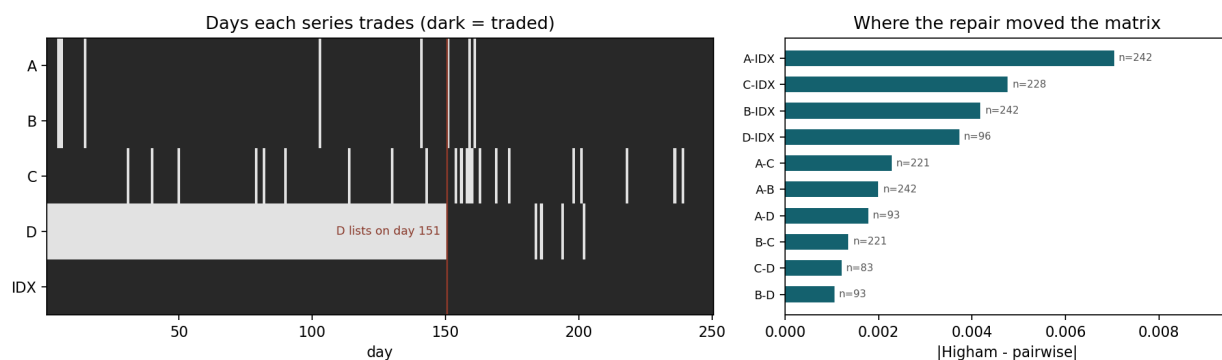


Figure 1: Left: the days each series trades. Right: how far the Higham repair moved each correlation, with the number of days behind it.

(a) Which estimator can return a matrix that is not PSD?

Pairwise can; complete case cannot. The complete case matrix is the correlation of one 81×5 data matrix X . Standardize its columns to get Z ; then $C = Z'Z/(n-1)$ and, for any vector w ,

$$w' C w = \frac{\|Zw\|^2}{n-1} \geq 0.$$

That is the definition of PSD, and it holds by construction. The pairwise matrix has no single X behind it. Entry (i, j) comes from the rows where i and j both trade, so the ten entries come from ten different samples, and nothing forces them to be consistent with each other. A set of inconsistent correlations is exactly what a negative eigenvalue is (Week 03).

(b) What does the near linear dependence of IDX imply?

Let $w = (-0.4, -0.3, -0.2, -0.1, 1)$ be long IDX and short the basket. Its variance is the tracking error variance, which is small. In correlation units the same portfolio is $v = w \circ \sigma / \|w \circ \sigma\|$, and

$$v' C v = \frac{\text{var}(w' r)}{\|w \circ \sigma\|^2} \geq \lambda_{\min}(C)$$

because the smallest eigenvalue is the minimum of $v' C v$ over unit vectors. The tracking variance can be measured directly on the 81 complete rows with no correlation estimate at all: 3.22×10^{-7} , a tracking error of 5.7 basis points a day. On the same rows $\|w \circ \sigma\|^2 = 2.11 \times 10^{-4}$, so **the smallest eigenvalue of the true matrix is at most about 0.0015**.

That number is the margin for error. An estimated matrix $\hat{C} = C + E$ has $v' \hat{C} v = v' C v + v' E v$, so any error with $v' E v < -0.0015$ in the tracking direction produces a negative eigenvalue. The pairwise entries carry sampling errors of 0.06 to 0.11 each (next part), and $v' E v$ sums ten of them weighted by $v_i v_j$, which reaches 0.36 for the A–IDX entry. **This data set is about as exposed as it could be. I expect the pairwise matrix to be non-PSD, with the negative eigenvector pointing along the tracking portfolio.**

(c) Which correlations are least reliable?

The four involving D, and C–D worst of all. A correlation's standard error is about $(1 - \rho^2)/\sqrt{n} \leq 1/\sqrt{n}$: 0.064 for anything among A, B and IDX, 0.10 for D's pairs, and 0.110 for C–D on 83 days. There is a second problem on top of the sample size. D's entries are estimated on days 151–250 only, while every other entry uses most of the 250 days. If the correlations changed over the year, D's row describes a different period from the rest of the matrix, and that is a source of inconsistency that no amount of data in the other entries can fix.

2.2 Fit

(d) Both matrices, their eigenvalues, and a Cholesky of each.

Table 2: Complete case (below the diagonal) and pairwise (above)

	A	B	C	D	IDX
A	1	0.4686	0.4309	0.4384	0.8696
B	0.3967	1	0.4630	0.2767	0.7329
C	0.2466	0.3169	1	0.1846	0.7136
D	0.4136	0.2361	0.1852	1	0.6057
IDX	0.8714	0.6709	0.5648	0.5719	1

Table 3: Eigenvalues, smallest first

	λ_1	λ_2	λ_3	λ_4	λ_5
Complete case	0.001506	0.5483	0.6933	0.8704	2.8864
Pairwise	-0.011072	0.4714	0.5284	0.8574	3.1539

- **Complete case: the Cholesky succeeds.** The smallest eigenvalue is 0.0015, just under the bound of 0.001525 from part (b), and positive.
- **Pairwise: the Cholesky fails,** in numpy and in the PSD tolerant `chol_psd`, at the last pivot, IDX, with a value of -0.0164 . A Cholesky pivot is the variance of a variable left over after regressing it on the variables before it, so the pairwise correlations say A through D explain 101.6% of the variance of IDX. That is the inconsistency, stated as a regression.

The eigenvector carrying the negative eigenvalue is $(-0.394, -0.234, -0.267, -0.209, 0.822)$, and the tracking portfolio in correlation units is $(-0.439, -0.238, -0.244, -0.152, 0.817)$. **The cosine between them is 0.997.** The prediction in (b) holds: the failure is the tracking portfolio.

(e) Tracking portfolio variance from the pairwise matrix.

Using each series' full history standard deviation (A 0.016502, B 0.011919, C 0.018324, D 0.022829, IDX 0.012281) and $\Sigma = DCD$:

$$w' \Sigma w = -1.80 \times 10^{-6}.$$

A negative variance. Measured directly on the complete rows, the same portfolio has a variance of $+3.22 \times 10^{-7}$.

(f) The two repairs.

Table 4: The two repairs of the pairwise matrix

Repair	Smallest eigenvalue	$\ \cdot - C_{pw}\ _F$	$w' \Sigma w$
Rebonato-Jackel	-7.5×10^{-17}	0.016182	6.81×10^{-7}
Higham	-9.9×10^{-10}	0.015520	6.79×10^{-7}

Both repairs set the negative eigenvalue to zero, up to rounding. Higham is 4% closer to the original, as it must be, since it is the nearest PSD correlation matrix in the Frobenius norm. The two repaired matrices are only 0.0046 apart. The tracking variance is now positive and about twice the direct measurement.

One practical consequence: the nearest PSD matrix to a non-PSD one sits on the boundary of the PSD set, so the repair is **singular**. numpy's Cholesky still fails on the Higham result, and `chol_psd` succeeds with 4 non-zero columns. To simulate from it you need the PSD Cholesky or a small ridge.

2.3 Reconcile

(g) Explain (e) with the definition of PSD.

Week 02 defines Σ as PSD when $w'\Sigma w \geq 0$ for every w . The reason that is the definition is that $w'\Sigma w$ is the variance of portfolio w , and no portfolio of real assets has a negative variance. The pairwise matrix fails the definition for exactly one direction, its negative eigenvector, and part (d) showed that direction is the tracking portfolio. So the matrix is not the covariance of any possible set of returns. It claims the hedge is better than perfect.

The data put the error in the worst place. The tracking portfolio is the one portfolio whose true variance is nearly zero, so it has almost no room for estimation error before going negative. It is also the portfolio a hedger actually holds.

(h) Which entries did Higham move most?

Table 5: Entries moved by the Higham repair, largest first. v is the negative eigenvector.

Pair	Days	Pairwise	Higham	Change	$v_i v_j$
A-IDX	242	0.8696	0.8625	-0.0070	-0.324
C-IDX	228	0.7136	0.7088	-0.0048	-0.219
B-IDX	242	0.7329	0.7287	-0.0042	-0.192
D-IDX	96	0.6057	0.6020	-0.0037	-0.171
A-C	221	0.4309	0.4332	+0.0023	+0.105
A-B	242	0.4686	0.4706	+0.0020	+0.092
A-D	93	0.4384	0.4402	+0.0018	+0.082
B-C	221	0.4630	0.4644	+0.0014	+0.062
C-D	83	0.1846	0.1858	+0.0012	+0.056
B-D	93	0.2767	0.2777	+0.0011	+0.049

The four IDX correlations, every one lowered. All six stock-stock correlations rose. The entries I named least reliable in (c), C-D and B-D, moved the *least* of all ten. Across the ten pairs the size of the change is negatively correlated with $1/n$ (-0.48): the repair moved the entries with more data behind them further.

That is because the repair was never told about the data. It sees only the matrix, and it fixes the matrix along its geometry. With one negative eigenvalue, clipping it changes entry (i, j) by about $-\lambda v_i v_j$, and the correlation between the size of each change and $|v_i v_j|$ is 1.0000. The eigenvector is dominated by IDX (loading 0.82), because IDX carries weight 1 in the tracking portfolio and no stock carries more than 0.4. So IDX's row absorbs the repair. Both directions of the change make sense: lowering IDX's correlations reduces how much of IDX the basket explains, and raising the stock correlations raises the basket's own variance. Both push the tracking variance back up through zero.

The repair is a statement about *where the inconsistency sits*, which is the tracking direction. It is not a statement about which entries are wrong. If the error is in D's row, as (c) suggested, the repair fixed it by changing IDX's row. Higham's weighted norm can be pointed at the less reliable entries, but only with weights the user supplies.

(i) Repair against estimator: which decision matters more?

$$\|C_{cc} - C_{pw}\|_F = 0.4239 \quad \|C_{\text{Higham}} - C_{pw}\|_F = 0.0155 \quad \|C_{\text{RJ}} - C_{\text{Higham}}\|_F = 0.0046$$

The choice of estimator matters 27 times more than whether to repair, and 90 times more than which repair. The largest gaps between the two estimators are all C’s correlations: A–C -0.184 , C–IDX -0.149 , B–C -0.146 .

The reason is that the complete case is also a choice of time window. Its 81 rows are all from days 151–250. Splitting the long-history pairs at day 151:

Table 6: Correlations before and after D lists

Pair	Days 1–150	Days 151–250	Fisher z
A–B	0.489	0.417	0.68
A–C	0.532	0.270	2.25
B–C	0.515	0.378	1.22
A–IDX	0.879	0.854	0.75
C–IDX	0.785	0.580	2.86

C’s correlations fell by a margin unlikely to be noise. The pairwise matrix mixes C’s high first half correlations into its C entries with D’s entries, which only exist in the low second half. That mixture of windows is a plausible source of the inconsistency in (d), and it is something no repair can see. So the first decision is which period the matrix is meant to describe. The repair is a 0.016 adjustment applied after that decision has already moved the matrix by 0.42.

3 A Volatility Estimate After the Regime Changed

Code: *problem2.py*.

3.1 Predict

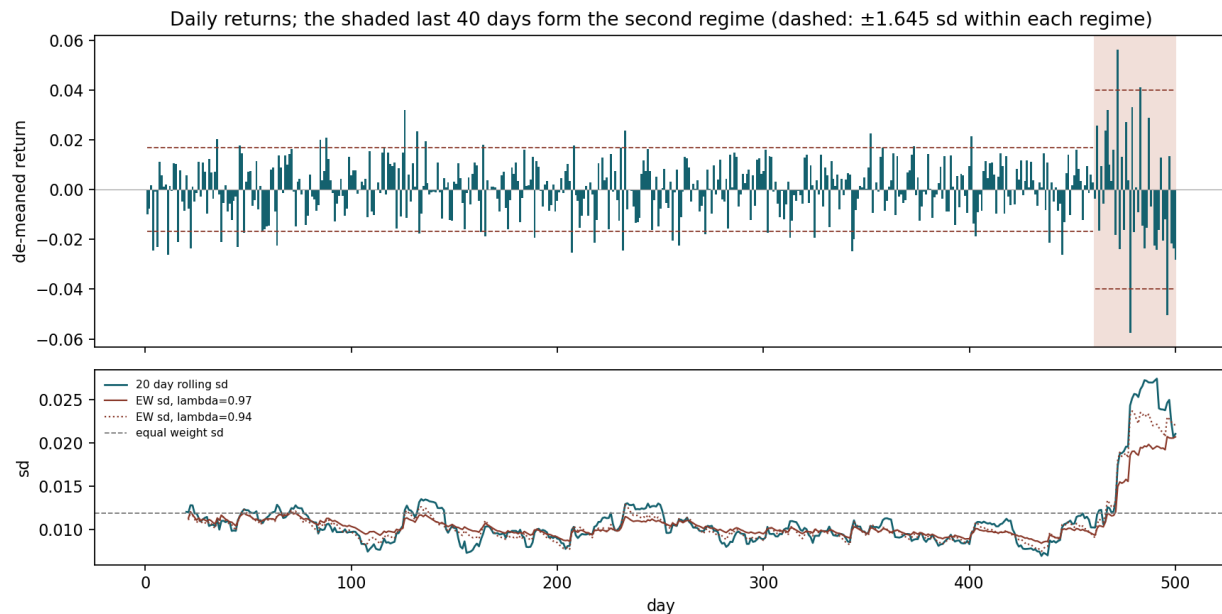


Figure 2: Top: the de-meaned returns, with the last 40 days shaded. Bottom: a 20 day rolling standard deviation, the two exponentially weighted estimates as they would have stood each day, and the equally weighted level.

The plot shows one feature: **the last 40 or so days swing about twice as wide as the 460 before them**, with no build up. The calm period looks the same from beginning to end.

(a) Rank the five VaR estimates before computing them.

My ranking, smallest to largest:

1. **Student's t .** It is fitted to all 500 days, so it sees one distribution whose tails are fat because two regimes are pooled. A fat tailed distribution fitted to the same data puts more mass in the centre and the far tails and less at the shoulders, and 5% is a shoulder. At the same variance its 5% quantile should sit slightly inside the normal's.
2. **Normal, equal weight.** The full sample standard deviation, 92% calm days.
3. **Historical.** The 5% quantile is the 25th worst of 500 days. The volatile days are 8% of the sample, and I expected them to crowd the worst 5% and push this just above the normal. Steps 1 to 3 should be close together.
4. **Normal, $\lambda = 0.97$.** Most of its weight is on the new regime (next part), so it should be near the new regime's level, but it still carries some calm days.
5. **Normal, $\lambda = 0.94$.** Almost all of its weight is on the new regime.

The first three are full sample estimates and the last two are recent ones, so I expected a wide gap between steps 3 and 4.

(b) Effective sample size, half life, and weight on the recent days.

Table 7: Exponential weights over $n = 500$ returns

λ	n_{eff}	Half life (days)	Weight on the last 40 days
0.97	65.67	22.76	0.704
0.94	32.33	11.20	0.916
equal	500	—	0.080

n_{eff} equals $(1 + \lambda)/(1 - \lambda)$ to two decimals, because λ^{500} is negligible and the finite weights behave like infinite ones. From the plot I put the regime change **40 days** before the end. A maximum likelihood changepoint scan, splitting the sample into two variances, puts it at 39, and the weights on 39 days (0.695 and 0.910) barely differ.

(c) Moments of the full sample. Could one normal regime produce them?

Mean zero (removed), standard deviation 0.011912, skewness -0.156 , excess kurtosis $+2.166$. Under normality the excess kurtosis has a standard error of about $\sqrt{24/n} = 0.219$, so 2.17 is **9.9 standard errors from zero**, and Jarque–Bera rejects at $p = 10^{-21}$. **No, one normal regime could not produce these.** My prediction is that two normal regimes with different variances can: a mixture of normals with different variances has positive excess kurtosis even when neither component has fat tails.

3.2 Fit

(d) The five VaRs on \$1,000,000.

Table 8: Five one day 5% VaR estimates, smallest first

Method	Standard deviation used	VaR
Student's t (MLE)	0.011818 (implied)	\$18,872
Normal, equal weight	0.011912	\$19,594
Historical simulation	—	\$20,574
Normal, EW $\lambda = 0.97$	0.020709	\$34,064
Normal, EW $\lambda = 0.94$	0.022074	\$36,308

Fitted t : location $\mu = 0.000182$, scale $\sigma = 0.010290$, $\nu = 8.270$. Its implied standard deviation is $\sigma\sqrt{\nu/(\nu - 2)} = 0.011818$.

3.3 Reconcile

(e) Compare with the ranking.

The order came out exactly as predicted, but one of the five landed there for a reason I had wrong.

The t sits below the normal for the reason given. Standardized to unit variance, the 5% quantile of a $t_{8,27}$ is -1.612 against the normal's -1.645 . The fitted location is also slightly positive, which removes another \$182.

The historical estimate is above the normal, as predicted, but **not because the volatile days crowd the tail**. Only 10 of the 25 worst days are from the last 40. The better way to check is to ask what historical simulation is estimating if the two regime picture is right. That is the 5% quantile of a 92/8 mixture of $N(0, 0.0102^2)$ and $N(0, 0.0243^2)$, which is **\$18,455, below the normal** and close to the t . The mixture is a fat tailed distribution, and at 5% a fat tailed distribution has the smaller quantile for the same reason the t does. The standard error of a 5% sample quantile from 500 days is about \$1,226, and the historical figure is 1.73 of those above the mixture value. So historical landing above the normal is sampling luck on the 25th order statistic. The correct prediction was “near the t ”, and I would have put it second, not third.

The two exponentially weighted estimates sit where predicted. The two regime picture says they should be $\sqrt{0.296s_1^2 + 0.704s_2^2} = 0.0211$ and $\sqrt{0.084s_1^2 + 0.916s_2^2} = 0.0234$, against the 0.0207 and 0.0221 actually estimated.

(f) Standard deviation in each regime, and what the t is describing.

Table 9: The two regimes

	Days	Standard deviation	Excess kurtosis	Normal VaR
Calm	460	0.010178	-0.20	\$16,741
Recent	40	0.024289	+0.03	\$39,952

Volatility rose by a factor of 2.39, and **inside each regime the returns are not fat tailed at all**. Both excess kurtoses are within about one standard error of zero. The excess kurtosis of the full sample comes entirely from pooling the two. A 92/8 mixture of two zero mean normals at these standard deviations has

$$\text{excess kurtosis} = \frac{3((1-p)s_1^4 + ps_2^4)}{((1-p)s_1^2 + ps_2^2)^2} - 3 = 2.57,$$

with a standard deviation of 0.011937. The sample has 2.17 and 0.011912.

The fitted t is describing the pooled, unconditional distribution of the last 500 days: a calm period and a volatile one, averaged into one fat tailed shape. Its standard deviation, 0.0118, matches neither regime. It answers “what does a randomly chosen day of the last two years look like” correctly, and “what does tomorrow look like” badly. Its VaR of \$18,872 is less than half the \$39,952 a normal VaR gives at the current regime’s volatility. Fitting a fatter tailed distribution to the full sample did not get closer to the current risk; it moved further away.

(g) Is the gap between the two EW VaRs larger than the noise?

Table 10: Estimation noise in each EW VaR

λ	$1/\sqrt{2n_{\text{eff}}}$	Standard error of the VaR
0.97	8.7%	\$2,972
0.94	12.4%	\$4,515

No. The gap is \$2,244, smaller than either standard error. Two caveats, which pull in opposite directions. The two estimates use the same days, so their errors are positively correlated and the noise in their *difference* is smaller than either standard error. But part of the gap is systematic rather than noise: $\lambda = 0.97$ still puts 30% of its weight on the calm regime, which the two regime picture says is worth \$3,799 by itself. Separately, the convention for the mean moves the numbers as much as λ does. Squaring around zero instead of the weighted mean gives \$38,707 at 0.94, \$2,399 higher, because the recent days have a negative weighted mean.

My opinion: **for a trading limit**, $\lambda = 0.94$. A limit exists to react, the change is 40 days old, and an estimator with a 23 day half life is still averaging in a regime that has ended. **For a capital number, 0.97**, as Week 03 suggests, because capital should not jump with every fortnight of data, and here it costs only 6.2% against 0.94. The more important point for capital is what not to use. The equally weighted, t , and historical numbers are all near \$19,000–21,000, half the current regime’s risk, and none of them should stand after a regime change this clear.

4 When Diversification Raises VaR

Code: *problem3.py*.

4.1 Predict

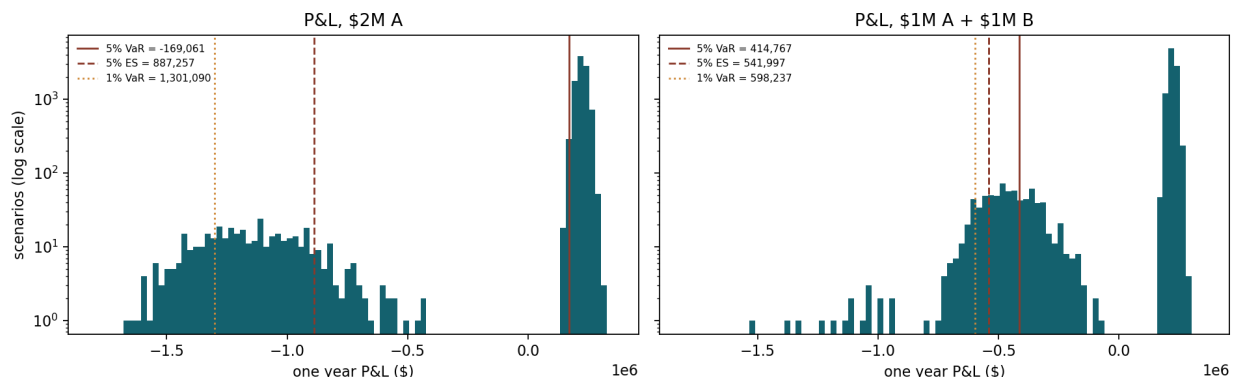


Figure 3: One year P&L of \$2,000,000 in A (left) and \$1,000,000 in each bond (right), log scale. Each bond's returns sit in two clusters, so the concentrated position has one loss cluster and the diversified one has a cluster for one default and a few scenarios where both default.

(a) Moments, and what they say about a normal VaR.

Table 11: Moments of each bond's one year return

Bond	Mean	SD	Skewness	Excess kurtosis	Min	Median	Max
A	0.0837	0.1360	-4.82	22.09	-0.830	0.1106	0.152
B	0.0850	0.1321	-4.92	23.09	-0.856	0.1105	0.154

These are the moments of a **two point mixture, not of anything bell shaped**. About 96% of scenarios sit tightly around +11.1%, the 100/90 - 1 a bond earns when it pays, and about 4% sit between -20% and -86%, the defaults. There is nothing in between: A's largest negative return is -0.219 and its smallest non-negative one is +0.069. The skewness of -4.8 is that 4% cluster far to the left, and the kurtosis of 22 is the distance between the clusters.

A normal VaR on these positions is meaningless. It uses only the mean and standard deviation, so it describes a single hump centred at 8.4% with a 13.6% spread. For \$1,000,000 in A it puts the 5% quantile at a loss of 14%, a region where not one of the 10,000 scenarios lies. It cannot know that the 5% point sits on a cliff edge.

(b) Counts, and a prediction for VaR and ES.

Table 12: Scenarios with a loss of more than 20%

Event	Scenarios	Share
A loses more than 20%	406	4.06%
B loses more than 20%	393	3.93%
At least one does	782	7.82%
Both do	17	0.17%
Both, if independent	16.0	

The defaults look independent: 17 joint against 16.0 expected, and a return correlation of 0.002. (One of B's 394 defaults lost only 19.6%, so the defaults number 406, 394 and 783. It changes nothing below.)

The 5% tail of 10,000 scenarios is the worst 500, and 500 sits between the counts.

- **VaR: the diversified choice has the larger VaR.** Concentrated in A, only 406 scenarios are defaults, fewer than 500, so the 500th worst scenario is a bond that paid. Its VaR will be *negative*, a profit. Diversified, 782 scenarios contain a default, more than 500, so the 500th worst contains one default and VaR will be a large loss, roughly $\$1,000,000$ times a default loss of 55% less the other bond's 11% gain.
- **ES: the diversified choice has the smaller ES.** ES averages the worst 500 scenarios, so it sees the defaults that VaR skips. Concentrated, 406 of the 500 are a default on the whole $\$2,000,000$. Diversified, all 500 contain a default, but almost all of them (483) are a default on half the money, offset by the other bond's gain. Roughly $0.81 \times \$2M \times 0.55$ against $\$1M \times (0.55 - 0.11)$: about $\$900,000$ against $\$450,000$.

4.2 Fit

(c) Historical VaR and ES at 5%, (d) normal VaR, (e) historical at 1%.

Table 13: VaR and ES of the four positions. The normal VaR uses each P&L's sample mean and standard deviation.

Position	VaR 5%	ES 5%	Normal VaR		ES 1%
			5%	VaR 1%	
\$1M in A	−\$84,531	\$443,628	\$139,971	\$650,545	\$709,548
\$1M in B	−\$86,011	\$419,587	\$132,277	\$641,248	\$700,805
\$2M in A	−\$169,061	\$887,257	\$279,941	\$1,301,090	\$1,419,096
\$1M in each	\$414,767	\$541,997	\$143,466	\$598,237	\$716,408

Both predictions hold. At 5%, VaR prefers the concentrated position by $\$583,828$, and ES prefers the diversified one by $\$345,260$.

4.3 Reconcile

(f) Subadditivity at 5%.

Table 14: $q(A) + q(B) \geq q(A + B)$, with the two \$1,000,000 positions

Measure	$q(A) + q(B)$	$q(A + B)$	Subadditive?
VaR 5%	−\$170,542	\$414,767	no
ES 5%	\$863,216	\$541,997	yes
VaR 1%	\$1,291,793	\$598,237	yes
ES 1%	\$1,410,353	\$716,408	yes
Normal VaR 5%	\$272,247	\$143,466	yes

VaR fails subadditivity at 5%; ES does not. Holding both bonds is measured as \$585,309 riskier than the sum of holding each.

The mechanism is the Week 04 bond example, at scale. VaR reads one point, the 500th worst scenario, and is blind to everything beyond it. Each bond on its own defaults in about 4% of scenarios, so its default sits entirely beyond the 5% point, where VaR cannot see it, and VaR reports a profit. Combining the bonds does not make any default more likely. It makes *some* default more likely: 7.8% of scenarios contain at least one, and that moves default mass across the 5% threshold. The worst 500 scenarios of the combined position are 483 single defaults and 17 double defaults. VaR then reports a single default in full. Nothing about the bonds changed; the quantile moved from one side of a cliff to the other.

ES is subadditive because it averages *every* scenario in the tail. It sees the 406 defaults in A's tail that VaR skipped, so the single bond ES figures are already large, and spreading the money over two independent bonds really does lower the average tail loss.

(g) The normal VaR says diversification helps. Is it right for the right reason?

It favours **the diversified position**, \$143,466 against \$279,941, and it is subadditive. Under normality VaR is 1.645σ less the mean, and σ is subadditive, so this outcome is guaranteed whatever the data look like. The correlation is 0.002, so the diversified standard deviation is close to $1/\sqrt{2}$ of the concentrated one, and that is the whole calculation.

It reaches the right choice for the wrong reason. The diversified position is the better one, because it has the lower ES and the lower 1% VaR. But every number the normal produces is wrong:

- it says the single bond VaR is a loss of \$139,971, when 95% of scenarios are a profit of at least \$84,531;
- it says the diversified VaR is \$143,466, when it is \$414,767, under by a factor of 2.9;
- it gets the ranking right only because it measures risk by standard deviation, which diversification always lowers. It has no idea where the defaults are.

A model that would have chosen the diversified position for *any* pair of uncorrelated assets has not discovered anything about these two.

(h) Why does the result change at 1%?

At 1% the tail is the worst 100 scenarios, and now every quantile sits beyond the default threshold. For \$2,000,000 in A, all 100 are defaults, and VaR is a loss of \$1,301,090. For the diversified position, the worst 100 are 83 single and 17 double defaults, and VaR is \$598,237, less than half. VaR is subadditive again, by a wide margin, and agrees with ES.

So VaR's failure shows up only when α falls between the probability of the bad event in the separate positions and its probability in the combined one, here between about 4% and 7.8%. Above 7.8%, no quantile sees a default. Below 4%, every quantile does. In between, diversification moves the event across the threshold and VaR punishes it. That makes the failure a property of the level chosen as much as of VaR, and the dangerous case is the one here: a rare, large loss whose probability sits just beyond the confidence level, which is exactly what credit portfolios and short option books look like.

5 Gaussian or t Copula

Code: *problem4.py*.

5.1 Predict

(a) Moments, and a margin for each series.

Table 15: Moments of each series, and the same with the most extreme day removed

Series	Mean	SD	Skewness	Excess kurtosis	Skewness, day 952 removed	Kurtosis, day 952 removed
X1	0.000257	0.011894	-0.034	-0.117	-0.006	-0.188
X2	-0.000252	0.016168	-0.615	7.125	+0.056	1.725
X3	0.000890	0.010019	0.074	2.874	+0.313	1.731

Standard errors under normality are about 0.077 for skewness and 0.155 for excess kurtosis at $m = 1000$.

- **X1: normal.** Both shape moments are within one standard error of zero.
- **X2: t , symmetric.** Its skewness of -0.62 and kurtosis of 7.1 look out of place. **Both rest on one day.** Day 952 is a return of -0.143 , an 8.8σ move. Without it the skewness is $+0.06$, gone, and the kurtosis falls to 1.72 , still 11 standard errors above zero. X2 is fat tailed, but not skewed, and not as fat tailed as the full moment says.
- **X3: t .** Excess kurtosis 2.87 , or 1.73 without its own extreme day.

Day 952 is the most extreme day of **all three series**, and each one's smallest return of the 1,000. One day on which every series has its worst day of the sample is the joint tail event this problem is about, before any copula is fitted.

(b) Ranks, joint tail counts, and a prediction.

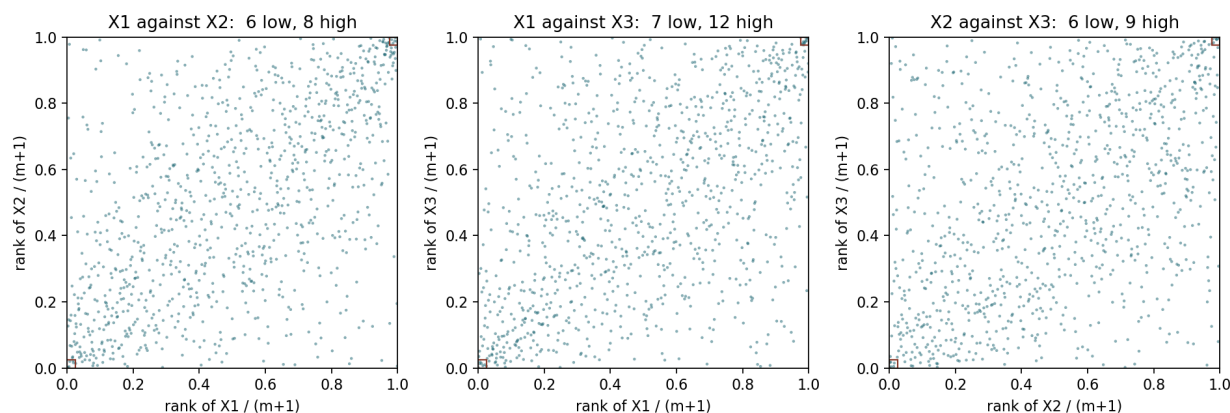


Figure 4: Each pair on ranks scaled into $(0,1)$. The boxes mark the joint worst 2.5% (bottom left) and joint best 2.5% (top right).

Table 16: Joint tail days out of 1,000

Pair	Kendall τ	$\sin(\pi\tau/2)$	Both in worst 2.5%	Both in best 2.5%	Gaussian at that ρ
X1–X2	0.396	0.583	6	8	5.93
X1–X3	0.335	0.503	7	12	4.66
X2–X3	0.307	0.464	6	9	4.14
Total			19	29	14.73

Under independence each pair would share $1000 \times 0.025^2 = 0.625$ **days per tail**. Every count is ten to twenty times that, but that only says the series are correlated, which the plots already show. The useful benchmark is a Gaussian copula at the same correlation: the last column is the expected count in each tail, computed from the bivariate normal at $\rho = \sin(\pi\tau/2)$. The data have **48 joint tail days against 29.5** expected under that Gaussian.

Is the tail more connected than the middle? Repeating the comparison at less extreme cut-offs:

Table 17: Joint exceedances relative to a Gaussian copula at the same correlation, three pairs and both tails

Cut-off q	25%	10%	5%	2.5%
Observed / Gaussian	1.04	1.11	1.17	1.63

The excess over the Gaussian **grows as the cut-off moves into the tail**: 4% at the quartiles and 63% at 2.5%. The plots show the same thing, a concentration of points in the two corners beyond what the elliptical cloud in the middle would suggest. Add day 952, and **I predict the t copula wins**, clearly.

5.2 Fit

(c) Normal or t for each margin, by AICc.

Table 18: Margin selection, $k = 2$ for the normal and 3 for the t

Series	ℓ normal	ℓ t	AICc		$\hat{\nu}$	Choice
			normal	AICc t		
X1	3013.274	3013.274	– 6022.54	–6020.52	> 1000	normal
X2	2706.253	2762.104	–5408.49	– 5518.18	5.27	t
X3	3184.851	3217.020	–6365.69	– 6428.02	6.09	t

As predicted. For X1 the t drives ν to infinity and gains nothing in likelihood, so the extra parameter loses on AICc by 2.01. X2’s t has $\nu = 5.27$, and refitted without day 952 it is 6.04: unlike the moments, the fitted t barely depends on the one observation.

(d) The copulas.

Each series goes to uniforms through its chosen margin. $R = \sin(\pi\tau/2)$ from Kendall's τ on those uniforms is the matrix in Table 16, since τ is a rank statistic. Its smallest eigenvalue is 0.413, so no repair was needed.

Table 19: Gaussian and t copula, same uniforms and same R , $m = 1000$

	Gaussian	t
Log likelihood ℓ	350.454	451.999
Free parameters k	0	1
$\hat{\nu}$	—	3.659
AICc	−700.91	−901.99
BIC	−700.91	−897.09

$$\Delta\text{BIC} = 2(\ell_t - \ell_{\text{Gauss}}) - \ln m = 203.09 - 6.91 = 196.18$$

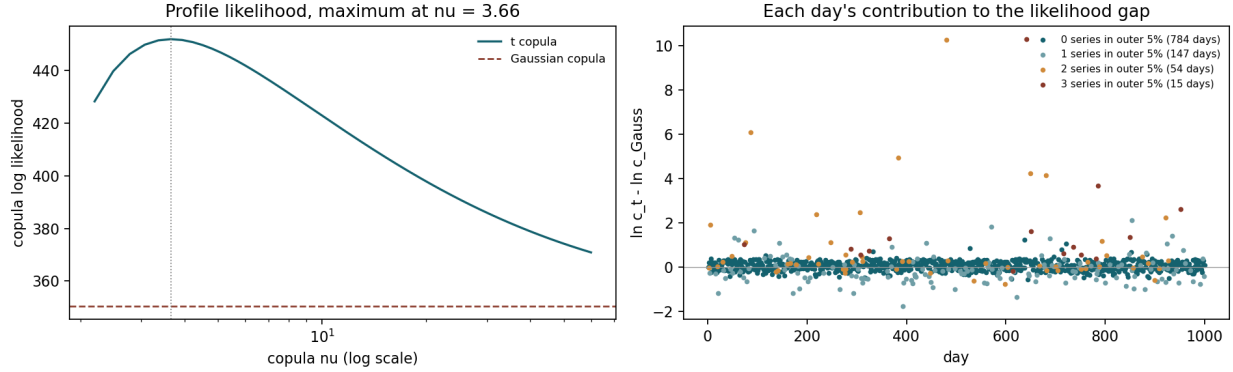


Figure 5: Left: the profile log likelihood of the t copula against ν , with the Gaussian level dashed. Right: each day's contribution to $\ell_t - \ell_{\text{Gauss}}$, coloured by how many series sit in their outer 5% that day.

(e) Portfolio VaR and ES, \$1,000,000 in each.

Table 20: Portfolio risk from 100,000 simulated days through the fitted margins, and historically

	VaR 5%	ES 5%	VaR 1%	ES 1%
Gaussian copula	\$49,477	\$66,218	\$75,971	\$93,561
t copula	\$48,656	\$67,924	\$77,981	\$101,412
Historical	\$47,939	\$65,699	\$70,092	\$102,424
t minus Gaussian	−\$821	+\$1,706	+\$2,010	+\$7,851
standard error of the difference	\$297	\$371	\$339	\$1,315

The two simulations share their normal draws, so the standard error of their difference is small. Each number on its own carries a simulation standard error of \$260–\$1,100.

(f) Joint tail days implied by each copula, per 1,000 days.

Table 21: Joint 2.5% tail days, observed and implied

Pair	Data, worst	Data, best	Gaussian, worst	Gaussian, best	t , worst	t , best
X1–X2	6	8	5.70	5.52	9.02	9.07
X1–X3	7	12	4.80	4.63	7.77	7.89
X2–X3	6	9	4.18	3.97	7.18	7.01
Total	19	29	14.68	14.12	23.97	23.97

5.3 Reconcile

(g) Which copula wins, and by how much?

The t , overwhelmingly: $\Delta\text{BIC} = 196$, $\Delta\text{AICc} = 201$. A BIC difference above 10 is already strong evidence. This one says the data are about $e^{101.5} \approx 10^{44}$ times more likely under the t . The profile in Figure 5 is sharply peaked at $\hat{\nu} = 3.66$, and the likelihood is within 1.92 units of its maximum only for ν from 3.10 to 4.43. Unlike the five stock example in Week 05, ν is well identified here, and it is low.

The joint tail counts agree. Over both tails the data have **48 days; the t implies 47.9 and the Gaussian 28.8**. Treating the count as Poisson, 48 or more has probability 0.0007 under the Gaussian and 0.52 under the t . (The three pairs share series, so the Poisson figure is approximate.)

One thing the t does not match is the split. The data have 19 joint worst days and 29 joint best, while the t copula is symmetric and implies 24 of each. A gap of 10 between two counts each near 24 is about 1.4 standard deviations, so it is not evidence of asymmetry on its own. If it persisted, an asymmetric family from the Archimedean group would be the next thing to try. Note too that the lower tail alone, 19, is closer to the Gaussian’s 14.7 than to the t ’s 24. It is the upper tail that most clearly rejects the Gaussian.

(h) Which risk number moves most, and why does 5% VaR barely move?

1% ES, up \$7,851 or 8.4%, six standard errors of the paired difference. 5% VaR moves by $-\$821$, -1.7% , and in the *opposite* direction: the t copula’s 5% VaR is slightly *lower*.

Both copulas use the same R and the same margins, so the portfolio standard deviation is the same under each (\$31,179 against \$31,196). What the t changes is where the probability sits for that variance. Its common shock puts more days in which all three move far together, and to keep the variance the same it pays for them with fewer moderately bad days. The 5% quantile sits near the point where those two effects cancel, which is the Week 05 point that a unit variance t has nearly the normal’s 5% quantile. Under the t copula the probability of losing more than the Gaussian’s 5% VaR is 4.78%, not 5%. Further out the picture reverses: the chance of exceeding the Gaussian’s 1% VaR is 1.12%, and the 0.1% loss quantile is \$131,572 against \$114,661, 15% higher. ES at 1% averages exactly the region where the copulas disagree most. VaR at 5% reads one point in the region where they agree.

The historical 1% ES, \$102,424, is within one standard error of the t copula’s \$101,412 and well above the Gaussian’s \$93,561. (The historical 1% VaR is the 10th worst of 1,000 days and too noisy to compare.)

(i) Where does $\ell_t - \ell_{\text{Gauss}}$ come from?

Table 22: Each day's $\ln c_t - \ln c_{\text{Gauss}}$, grouped by how many series sit in their outer 5% on either side (by rank)

Series in their outer 5%	Days	Contribution	Share	Per day
0	784	+49.84	49.1%	+0.064
1	147	−19.02	−18.7%	−0.129
2	54	+44.44	43.8%	+0.823
3	15	+26.28	25.9%	+1.752
Total	1,000	+101.55		

Half the total, 49.8 of 101.5, comes from the 784 days on which no series is in its outer 5%. Each such day contributes little, 0.064, but there are many of them. And within those days the gain is concentrated near the centre: the half of them closest to the middle of the copula contributes +48.2, the outer half +1.7. The t copula with the same R is more peaked in the centre than the Gaussian, and the data are too. At the other end, the 147 days on which exactly one series is extreme count *against* the t by 19.0, because the t says extremes come together and a lone extreme is less likely under it. Day 952 on its own contributes only 2.6.

The information criterion is scoring the whole shape of the dependence, and by number of days that is mostly the middle. It is not a tail test. Here the middle and the tails happen to agree, and part (f) confirmed the tails separately. But a copula could win on AICc by fitting the peak in the centre and still get the joint tail wrong, and the criterion would not say so. The tail counts are the check the criterion cannot do. This is the Week 05 point that almost all of the sample sits in the middle; here it can be measured, and it is half.

(j) Tail dependence for the most correlated pair.

For X1–X2, $\rho = 0.583$ and $\hat{\nu} = 3.66$:

$$\lambda = 2 t_{\nu+1} \left(-\sqrt{\frac{(\nu+1)(1-\rho)}{1+\rho}} \right) = 0.322$$

Under the Gaussian copula it is 0, as it is for every $\rho < 1$. The t copula says that given X1 has a loss deep in its tail, there is about a one in three chance that X2 does too, however deep the tail. Across the likelihood range for ν in (g), λ runs from 0.281 to 0.357. The other two pairs have 0.273 and 0.253.

6 Model Based Simulation and Residual Correlation

Code: *problem5.py*.

6.1 Predict

(a) Does treating ϵ_A and ϵ_B as independent overstate or understate the VaR of each portfolio?

With $r_i = \alpha_i + \beta_i r_M + \epsilon_i$, the market uncorrelated with each error, and h dollars a side, the Week 02 portfolio variance formula gives

$$\text{var}(P1) = h^2 [(\beta_A + \beta_B)^2 \sigma_M^2 + \sigma_A^2 + \sigma_B^2 + 2\rho \sigma_A \sigma_B]$$

$$\text{var}(P2) = h^2 [(\beta_A - \beta_B)^2 \sigma_M^2 + \sigma_A^2 + \sigma_B^2 - 2\rho \sigma_A \sigma_B]$$

where σ_A, σ_B are the residual standard deviations and ρ is the residual correlation. A shared industry the market does not capture shows up as $\rho > 0$. The shortcut sets $\rho = 0$, which:

- removes a **positive** term from P1, so it **understates** P1's VaR;
- removes a **negative** term from P2, so it **overstates** P2's VaR.

I also expect P2 to be the more exposed. If the two betas are similar, the market term nearly cancels in P2, so the residual terms are almost all of its variance, and the covariance term is a large fraction of that.

6.2 Fit

(b) The two regressions.

Table 23: OLS of each stock's daily arithmetic return on the market, $n = 500$

Stock	α	β	Residual SD ($n - 2$)	Residual SD ($n - 1$)	R^2
A	0.000205	1.0353	0.012901	0.012888	0.418
B	0.000315	0.9354	0.011550	0.011538	0.423

The residual correlation is 0.698. The market explains only about 42% of each stock's variance, and what it leaves behind is strongly shared. That is the shared industry. The market's standard deviation is 0.010553.

(c) **Simulated VaR under the two assumptions**, with $[r_M, \epsilon_A, \epsilon_B] \sim N(0, \Sigma)$ and Σ block diagonal, the Week 05 construction. Expected returns are zero, so the alphas are dropped. Both runs use the same random numbers. All covariances use the $n - 1$ divisor, to match (d).

Table 24: One day 5% VaR from 100,000 simulated days

Residual covariance	P1 VaR	P2 VaR	P1 ES	P2 ES
Full	\$50,298	\$15,841	\$63,152	\$19,876
Off diagonal set to 0	\$44,450	\$28,472	\$55,712	\$35,725

(d) **Delta normal VaR** from the sample covariance of A and B: **P1 \$50,415, P2 \$15,834.**

6.3 Reconcile

(e) **Did the shortcut move each portfolio in the predicted direction?**

Yes, both. P1 falls from \$50,298 to \$44,450, -11.6% , understated. P2 rises from \$15,841 to \$28,472, $+79.7\%$, overstated. P2 is far more exposed. The variance decomposition shows why:

Table 25: Each portfolio's daily variance in millions of dollars squared

	Market	Residual variances	Residual covariance	Total	Covariance share
P1	432.5	299.2	+207.7	939.4	+22.1%
P2	1.1	299.2	-207.7	92.7	- 224.1%

The betas are 1.035 and 0.935, so P2's net beta is 0.100 and its market term is almost nothing. What is left is the two residual variances, *less* the covariance term, which hedges away 69% of them. Deleting that term makes P2's variance 3.24 times larger and its VaR $\sqrt{3.24} = 1.80$ times larger. For P1 the covariance term is a fifth of a variance that is mostly market, so deleting it costs 12%. A long-short portfolio of two names in the same industry is a bet on their residual difference, so the residual correlation is the thing it depends on most, and that is exactly what the shortcut throws away.

(f) **Full residual covariance against delta normal.**

They are the same number. In closed form the model VaR is \$50,415 and \$15,834, identical to the delta normal VaR. The model implied covariance $\beta\beta'\sigma_M^2 + \Sigma_\epsilon$ matches the sample covariance of A and B to 1.6×10^{-19} . The simulated \$50,298 and \$15,841 differ only by simulation error.

This is the Week 05 statement that OLS with normal errors *is* the joint normal. Writing $r = \beta r_M + \epsilon$,

$$\text{cov}(r) = \beta\beta'\sigma_M^2 + \beta\text{cov}(r_M, \epsilon)' + \text{cov}(r_M, \epsilon)\beta' + \Sigma_\epsilon,$$

and OLS makes the residuals exactly uncorrelated with the regressor *in sample* (here -5.8×10^{-20}). So the middle terms vanish and the regression returns the sample covariance, reorganized into a market part and a residual part. Fitting the model and simulating its pieces gives back the joint normal you could have written down at the start. The model only changes the answer when it imposes something, and the diagonal Σ_ϵ in (c) is exactly such a restriction. In that case it was a wrong one.

The two stop agreeing when any of the conditions behind that identity breaks:

- **Non-normal pieces.** t distributed errors or a fat tailed market, as in the Week 05 example. The simulated P&L is then not normal, and its quantile is not 1.645σ , even if the covariance still matches.
- **A restricted Σ_ϵ ,** as in the shortcut, or any factor model that assumes the residuals are uncorrelated.
- **Pieces estimated on different samples or weights,** for example an exponentially weighted market variance with an equally weighted residual covariance, or betas from a longer history. Then $\beta\beta'\sigma_M^2 + \Sigma_\epsilon$ is no longer the sample covariance of any one window.
- **Betas not from OLS,** such as a t regression or a robust fit. Its residuals are not exactly orthogonal to the market in sample.
- **Non-linear positions.** The delta normal linearizes, and the simulation can reprice.
- **Divisor conventions,** trivially. Using the regression's $n - 2$ residual variance rather than $n - 1$ moves the model VaR in the fourth significant figure.